



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education

CHEMISTRY

0620/21

Paper 2 Multiple Choice (Extended)

October/November 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

A copy of the Periodic Table is printed on page 16.

Electronic calculators may be used.

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **13** printed pages and **3** blank pages.



- 1 Samples of four gases are released in a room at the same time.

The gases are carbon dioxide, CO_2 , hydrogen chloride, HCl , hydrogen sulfide, H_2S , and nitrogen dioxide, NO_2 .

Which gas diffuses fastest?

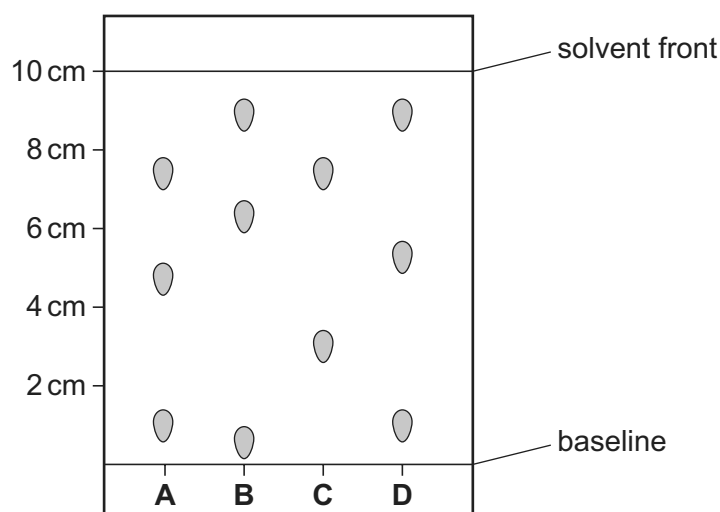
- A carbon dioxide
 - B hydrogen chloride
 - C hydrogen sulfide
 - D nitrogen dioxide
- 2 A student is asked to measure the time taken for 0.4 g of magnesium carbonate to react completely with 25.0 cm^3 of dilute hydrochloric acid.

Which pieces of apparatus does the student need?

- A balance, stop-clock, pipette
 - B balance, stop-clock, thermometer
 - C balance, pipette, thermometer
 - D stop-clock, pipette, thermometer
- 3 Four different food colourings are analysed using chromatography.

The results are shown on the chromatogram. The diagram is not drawn to scale.

Which food colouring contains a component with an R_f value of 0.3?



4 Which statement about an ionic compound is **not** correct?

- A It conducts electricity when dissolved in water.
- B It has a high melting point due to strong attractive forces between ions.
- C It has a regular lattice of oppositely charged ions in a 'sea of electrons'.
- D The ionic bonds are formed between metallic and non-metallic elements.

5 An isotope of chromium is represented by ${}^{52}_{24}\text{Cr}$.

Which statement about an atom of this isotope of chromium is correct?

- A It contains 24 electrons.
- B It contains 24 neutrons.
- C It contains 28 protons.
- D It contains 52 neutrons.

6 Element X has two isotopes, ${}^{12}_6\text{X}$ and ${}^{14}_6\text{X}$.

Which statement about these isotopes is correct?

- A They have different chemical properties because they have different numbers of neutrons.
- B They have the same chemical properties because they have the same number of outer shell electrons.
- C They have the same nucleon number because the sum of the number of protons and electrons is the same.
- D They have different positions in the Periodic Table because they have different numbers of neutrons.

7 How are the structures of diamond and silicon(IV) oxide similar?

- A Molecules of both diamond and silicon(IV) oxide are held together by weak attractive forces.
- B They both contain atoms arranged in planes held together by weak bonds.
- C They both contain ions that are free to move.
- D The carbon in diamond and the silicon in silicon(IV) oxide each have four covalent bonds.

8 Which statement describes the structure of copper?

- A It has a lattice of negative ions in a 'sea of electrons'.
- B It has a lattice of negative ions in a 'sea of protons'.
- C It has a lattice of positive ions in a 'sea of electrons'.
- D It has a lattice of positive ions in a 'sea of protons'.

- 9 Four fertilisers are each supplied in 100 kg bags.

Which fertiliser supplies the greatest mass of nitrogen per 100 kg bag?

- A ammonium nitrate, NH_4NO_3
 B ammonium phosphate, $(\text{NH}_4)_3\text{PO}_4$
 C ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$
 D urea, $\text{CO}(\text{NH}_2)_2$

- 10 Calcium carbonate reacts with dilute hydrochloric acid.

The equation for the reaction is shown.



1.00 g of calcium carbonate is added to 50.0 cm^3 of 0.0500 mol/dm^3 hydrochloric acid.

Which volume of carbon dioxide is made in this reaction?

- A 30 cm^3 B 60 cm^3 C 120 cm^3 D 240 cm^3

- 11 Which rows correctly show cathode and anode products from the electrolysis of the named electrolyte?

	electrolyte	cathode product	anode product
1	copper(II) sulfate solution using copper electrodes	copper	oxygen
2	molten lead(II) bromide	lead	bromine
3	dilute sodium bromide solution	hydrogen	oxygen
4	copper(II) sulfate solution using carbon electrodes	hydrogen	oxygen

- A 1 and 2 only B 1 and 4 only C 2 and 3 only D 3 and 4 only

- 12 What are the ionic half-equations for the electrode reactions during the electrolysis of concentrated aqueous sodium chloride?

	anode	cathode
A	$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
B	$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
C	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$
D	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

13 Which statements about endothermic reactions are correct?

- 1 The energy of the products is greater than the energy of the reactants.
- 2 The energy of the reactants is greater than the energy of the products.
- 3 The temperature of the surroundings increases during the reaction.
- 4 The temperature of the surroundings decreases during the reaction.

A 1 and 3 only **B** 1 and 4 only **C** 2 and 3 only **D** 2 and 4 only

14 Which gases are used to generate electricity in a fuel cell?

- A** carbon dioxide and oxygen
B hydrogen and methane
C hydrogen and oxygen
D methane and carbon dioxide

15 Which is a chemical change?

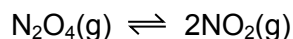
- A** boiling water
B cooking an egg
C dissolving sugar
D melting ice cubes

16 The rate of reaction between magnesium and dilute hydrochloric acid is increased by increasing the concentration of the acid.

How does this affect the reacting particles?

	collision rate of particles	proportion of particles with sufficient energy to react
A	increases	increases
B	increases	stays the same
C	stays the same	increases
D	stays the same	stays the same

- 17 Dinitrogen tetroxide, N_2O_4 , is converted into nitrogen dioxide, NO_2 , in a reversible reaction.

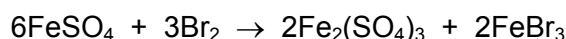


The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

	pressure /atmospheres	temperature
A	2	high
B	2	low
C	50	high
D	50	low

- 18 The equation for the reaction between iron(II) sulfate and bromine is shown.



Which row identifies the oxidising agent and the reducing agent?

	oxidising agent	reducing agent
A	Br_2	FeSO_4
B	FeSO_4	Br_2
C	FeBr_3	$\text{Fe}_2(\text{SO}_4)_3$
D	$\text{Fe}_2(\text{SO}_4)_3$	FeBr_3

- 19 Which statement about amphoteric oxides is correct?

- A** They are made by combining an acidic oxide with a basic oxide.
- B** They react with water to give a solution of pH 7.
- C** They react with both acids and bases.
- D** They do not react with acids or bases.

- 20 Carbonic acid is a weak acid formed when carbon dioxide dissolves in water.

What is the pH of the solution?

- A** 1 **B** 5 **C** 7 **D** 9

21 A method used to make copper(II) sulfate crystals is shown.

- 1 Place dilute sulfuric acid in a beaker.
- 2 Warm the acid.
- 3 Add copper(II) oxide until it is in excess.
- 4 Filter the mixture.
- 5 Evaporate the filtrate until crystals start to form.
- 6 Leave the filtrate to cool.

What are the purposes of step 3 and step 4?

	step 3	step 4
A	to ensure all of the acid has reacted	to obtain solid copper(II) sulfate
B	to ensure all of the acid has reacted	to remove the excess of copper(II) oxide
C	to speed up the reaction	to obtain solid copper(II) sulfate
D	to speed up the reaction	to remove the excess of copper(II) oxide

22 Lead(II) sulfate is an insoluble salt.

Which process is **not** used to prepare a pure sample of this salt?

- A** crystallisation
B drying
C filtration
D precipitation

23 Part of the Periodic Table is shown.

Which element is used to provide an inert atmosphere?

A simplified periodic table with 18 columns and 4 rows. The columns are grouped into four sets of three. Element A is in the top row, middle column. Element B is in the second row, second column of the third group. Element C is in the second row, third column of the fourth group. Element D is in the third row, third column of the second group.

- 24 Which pair of elements reacts together most violently?
- A chlorine and lithium
 - B chlorine and potassium
 - C iodine and lithium
 - D iodine and potassium
- 25 Which pair of compounds shows that transition elements have variable oxidation states?
- A Cr_2O_3 and CrBr_3
 - B CuSO_4 and CuCl_2
 - C Fe_2O_3 and FeCl_2
 - D NiO and NiCl_2
- 26 Some properties of substance X are listed.
- It conducts electricity when molten.
 - It has a high melting point.
 - It burns in oxygen and the oxide dissolves in water to give a solution with pH 11.

What is X?

- A a covalent compound
 - B a macromolecule
 - C a metal
 - D an ionic compound
- 27 Which statement is correct?
- A Aluminium is used in the manufacture of aircraft because it has a high density.
 - B Copper is used for cooking utensils because it is a good conductor of heat.
 - C Mild steel is used for car bodies because it is resistant to corrosion.
 - D Stainless steel is used for cutlery because it is a conductor of electricity.

28 Iron rusts but aluminium does not easily corrode.

Which statement explains why aluminium does **not** easily corrode?

- A** It is an alloy.
- B** It is below iron in the reactivity series.
- C** It is not a transition element.
- D** Its surface is protected by an oxide layer.

29 Which statement about the extraction of aluminium is correct?

- A** Aluminium is formed at the cathode during the electrolysis of aluminium oxide.
- B** Hematite is mainly aluminium oxide.
- C** Molten cryolite is used to raise the melting point of the aluminium oxide.
- D** Oxygen gains electrons at the anode during the electrolysis of aluminium oxide.

30 River water contains soluble impurities, insoluble impurities and bacteria.

River water is made safe to drink by filtration and chlorination.

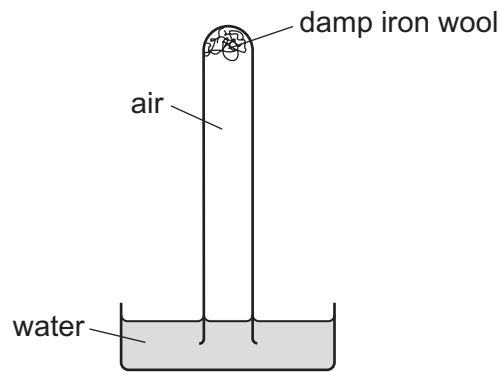
Which statement is correct?

- A** Filtration removes bacteria and insoluble impurities, and chlorination removes soluble impurities.
- B** Filtration removes insoluble impurities, and chlorination kills the bacteria.
- C** Filtration removes soluble and insoluble impurities, and chlorination kills the bacteria.
- D** Filtration removes soluble impurities and bacteria, and chlorination removes insoluble impurities.

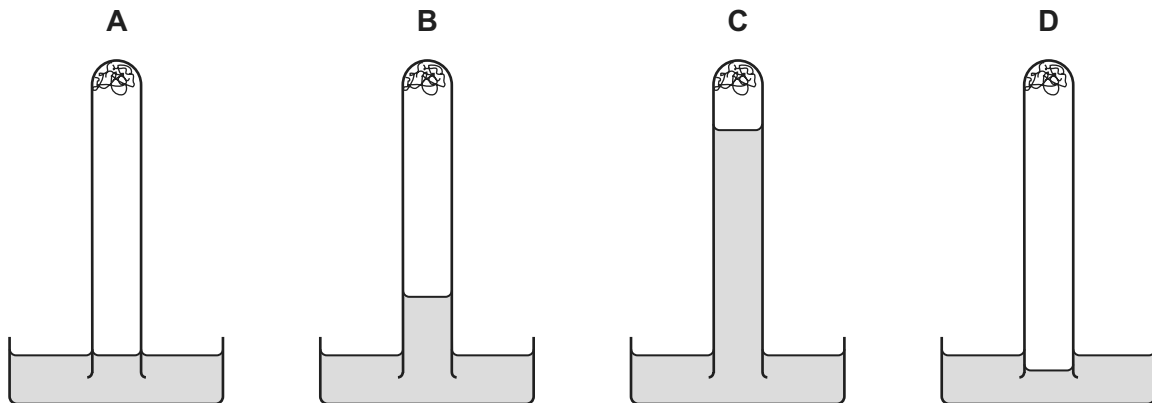
31 Which physical property is used to separate the nitrogen and oxygen from air?

- A** boiling point
- B** density
- C** electrical conductivity
- D** molecular mass

32 The apparatus shown is set up and left for a week.



Which diagram shows the level of the water at the end of the week?



33 Which statement about the carbon cycle is correct?

- A** Carbon is absorbed from the atmosphere by combustion and released into it by respiration.
- B** Carbon is absorbed from the atmosphere by photosynthesis and released into it by combustion.
- C** Carbon is absorbed from the atmosphere by both respiration and combustion.
- D** Carbon is released into the atmosphere by both photosynthesis and respiration.

34 Ammonium sulfate is used as a fertiliser.

It is made from ammonia and sulfuric acid.

Which words complete gaps 1, 2 and 3?

The1..... is made by the2..... process in which3..... is used as a catalyst.

	1	2	3
A	ammonia	Contact	iron
B	ammonia	Haber	vanadium(V) oxide
C	sulfuric acid	Contact	vanadium(V) oxide
D	sulfuric acid	Haber	iron

35 Which process is used to obtain lime from limestone?

- A** cracking
- B** fractional distillation
- C** neutralisation
- D** thermal decomposition

36 Petroleum is separated by fractional distillation.

Which statement about the fractions produced is correct?

- A** Bottled gas for heating and cooking is obtained from the naphtha fraction.
- B** Diesel oil is used as a fuel for jet aircraft.
- C** Substances used to make polishes are obtained from the lubricating fraction.
- D** The kerosene fraction contains many useful waxes.

37 Which products are obtained by the cracking of an alkane?

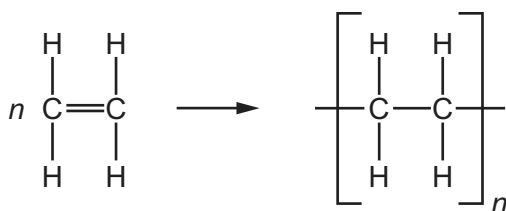
	alkene	hydrogen	water
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	✓	✓

38 Ethanol is manufactured by the catalytic addition of steam to ethene and by fermentation.

Which statement describes an advantage of fermentation compared to the catalytic addition of steam to ethene?

- A** Fermentation is a more rapid reaction.
- B** Fermentation produces a purer product.
- C** Fermentation uses a higher temperature.
- D** Fermentation uses renewable resources.

39 The diagram shows the structure of a monomer and of the polymer made from it.



What are the monomer and polymer?

	monomer	polymer
A	ethane	poly(ethane)
B	ethane	poly(ethene)
C	ethene	poly(ethane)
D	ethene	poly(ethene)

40 Which polymers possess the same linkage?

- A** nylon and protein
- B** protein and starch
- C** starch and nylon
- D** nylon and *Terylene*

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
		1Hhydrogen1																	
		Key atomic number atomic symbol name relative atomic mass																	
3Li lithium 7	4Be beryllium 9													5B boron 11	6C carbon 12	7N nitrogen 14	8O oxygen 16	9F fluorine 19	10Ne neon 20
11Na sodium 23	12Mg magnesium 24													13Al aluminium 27	14Si silicon 28	15P phosphorus 31	16S sulfur 32	17Cl chlorine 35.5	18Ar argon 40
19K potassium 39	20Ca calcium 40	21Sc scandium 45	22Ti titanium 48	23V vanadium 51	24Cr chromium 52	25Mn manganese 55	26Fe iron 56	27Co cobalt 59	28Ni nickel 59	29Cu copper 64	30Zn zinc 65	31Ga gallium 70	32Ge germanium 73	33As arsenic 75	34Se selenium 79	35Br bromine 80	36Kr krypton 84		
37Rb rubidium 85	38Sr strontium 88	39Y yttrium 89	40Zr zirconium 91	41Nb niobium 93	42Mo molybdenum 96	43Tc technetium —	44Ru ruthenium 101	45Rh rhodium 103	46Pd palladium 106	47Ag silver 108	48Cd cadmium 112	49In indium 115	50Sn tin 119	51Sb antimony 122	52Te tellurium 128	53I iodine 127	54Xe xenon 131		
55Cs caesium 133	56Ba barium 137	57–71lanthanoids	72Hf hafnium 178	73Ta tantalum 181	74W tungsten 184	75Re rhenium 186	76Os osmium 190	77Ir iridium 192	78Pt platinum 195	79Au gold 197	80Hg mercury 201	81Tl thallium 204	82Pb lead 207	83Bi bismuth 209	84Po polonium —	85At astatine —	86Rn radon —		
87Fr francium —	88Ra radium —	89–103actinoids	104Rf rutherfordium —	105Db dubnium —	106Sg seaborgium —	107Bh bohrium —	108Hs hassium —	109Mt meitnerium —	110Ds darmstadtium —	111Rg roentgenium —	112Cn copernicium —	113Nh nihonium —	114Fl flerovium —	115Lv livermorium —	116Ts tennessine —	117Og oganeson —	118—		

lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).